

# A PERFECT AUTOPILOT FOR PACKAGE DRONE DELIVERY

UAV Navigation
Cutting-edge Autopilots

# POST-COVID AVIATION INDUSTRY

COVID-19 pandemic has been one of the hardest setbacks the aviation industry has suffered in history. The reduction of people's movement significantly impacted global air traffic. In 2021, there have been green shoots as the number of flights has increased to figures close to pre-pandemic figures.

In the civil context, the global drone market is expected to reach US$41.3B in 2026 with a Compound annual growth rate CAGR of 9.4% [1]. In this segment, the most demanded functionalities [2] are Urban Air Mobility (UAM), which Deloitte estimates a value of US$115B in 2035 [3], and the packaging delivery using drones, with a market growth rate of 48% by 2029 [4].

The situation during the COVID pandemic has shown the critical role that unmanned aviation may play in developing a more efficient and safer world. The Spanish Aerospace Technology Program (CDTI) [5] is considered an excellent opportunity to make the sector grow in the following years, **creating a more sustainable and green industry**.

> Drones are a clear part of the future transport and logistic landscape
>
> Adina Vălean
> European Commissioner for Transport [6]



# ENABLING DRONES IN LOGISTIC

Following the industry's evolution, UAV Navigation has improved its delivery capabilities in the flight control solution to **enable its clients to offer package delivery services and allow the use of their drones in logistics**. In this sense, we have refined important functionalities to make this capability more accessible, safe, and easy to use.

# AUTOMATIC ACTIONS ON WAYPOINTS

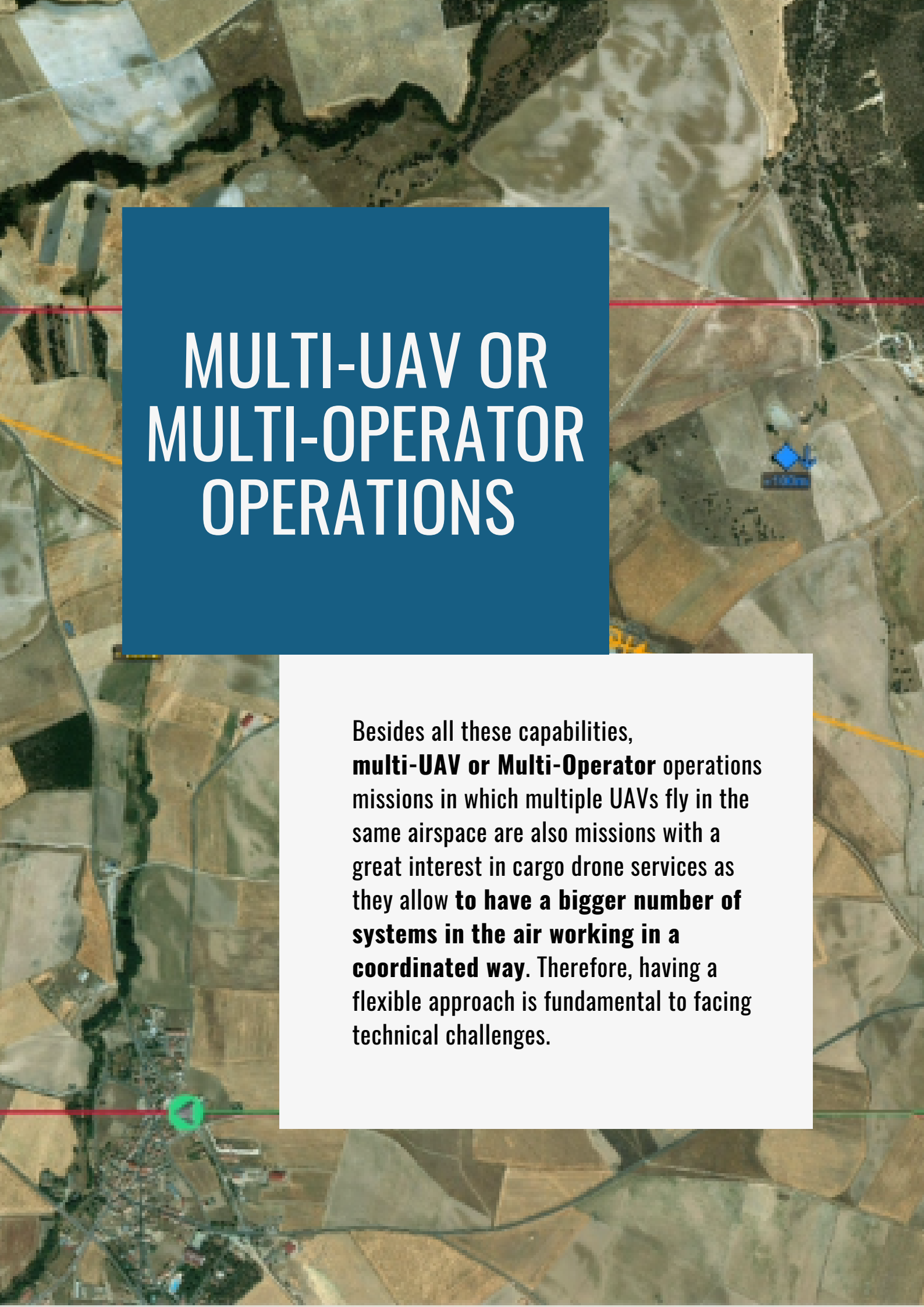
One of these features is the automatic actions on waypoints, allowing the platform to perform a complete delivery service without operator intervention. The package may be delivered automatically via drone while the staff is preparing another package to be delivered after that, reducing costs and time-to-client. This methodology would optimize and even resolve the famous Last-Mile Delivery problem. These services require an advanced and reliable system with logic that increases flight safety. UAV Navigation system includes configurable conditional actions following the alarms that may appear during the flight. The autopilot may change its course towards an emergency or alternative aerodrome and complete the mission in case of lack of fuel, comm loss, or other circumstances and ensuring the aircraft and the package integrity.

# UAV NAVIGATION POLAR AHRS AND VTOL SOLUTION

Our expertise in unmanned aviation and the advanced algorithms included in our AHRS help us have extraordinary control during landing without any external infrastructure, which eases commercial launch and reduces the associated costs of the cargo delivery service. The POLAR AHRS, included in our VECTOR range of autopilots, uses an ADS system to estimate the wind direction and perform a safe delivery of the package.

This technique is also used in our flight control solution for VTOL platforms, where the aircraft's heading may be critical in their vertical takeoff and landing stages of the flight. These kinds of platforms are also beneficial in delivery missions as they can cover long distances in a short period and perform a hover maneuver to execute the cargo drop. UAV Navigation flight control system may control hybrid UAV VTOL platforms, creating flight plans for a fully automatic and autonomous operation of these aircraft configurations possible.





# MULTI-UAV OR MULTI-OPERATOR OPERATIONS

Besides all these capabilities, **multi-UAV or Multi-Operator** operations missions in which multiple UAVs fly in the same airspace are also missions with a great interest in cargo drone services as they allow **to have a bigger number of systems in the air working in a coordinated way**. Therefore, having a flexible approach is fundamental to facing technical challenges.

# PROJECT VANET

Considering the necessities and requirements of this industry, apart from the previous capabilities and features, we have presented, together with CATEC, the project VaNeT within the CERVERA call. This project aims to develop and promote a multi-agent network node for safe freight transport by UAVs in shared air spaces. This project is funded by the Center for Technological and Industrial Development (CDTI) and the Spanish Ministry of Science and Innovation. It is also co-financed by the European Regional Development Fund (FEDER).

UAV Navigation keeps working day-by-day to set the path of future drone applications and help the industry evolve and grow. For this purpose, we keep on developing new tools that allow executing more complex and demanding operations in a safe and efficient way, which helps improve the operator's situational awareness.



# References

[1] HTTPS://DRONEII.COM/THE-DRONE-MARKET-IN-2021-AND-BEYOND-5-KEY-TAKEAWAYS
[2] HTTPS://DRONEII.COM/PRODUCT/DRONE-MARKET-REPORT. CONSULTADO EL 16 OCTUBRE 2021
[3] HTTPS://WWW2.DELOITTE.COM/GLOBAL/EN/PAGES/ABOUT-DELOITTE/ARTICLES/COVID-19/UNDERSTANDING-COVID-19-IMPACT-ON-AEROSPACE-AND-DEFENSE.HTML
[4] HTTPS://WWW.UASVISION.COM/2021/09/23/PACKAGE-DRONES-WORLD-MARKET-TO-GROW-48-BY-2029/. CONSULTADO EL 16 OCTUBRE 2021
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[6] HTTPS://WWW.EUROCONTROL.INT/ARTICLE/EUROPE-NOW-FAST-LANE-IMPLEMENTING-UAS-TRAFFIC-MANAGEMENT-SYSTEMS. CONSULTADO EL 16 OCTUBRE 2021.

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WWW.UAVNAVIGATION.COM


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